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Optimization of Machining Parameters for Turning of Al6061 using Robust Design Principle to minimize the surface roughness

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Abstract

The present work aims at optimization of the process parameter such as cutting speed, depth of cut and feed rate on surface roughness produced on the machined component. Analysis is carried out using Taguchi robust design principles. From the analysis, feed rate is found to be the most influential process parameters which influence the surface roughness followed by cutting speed and depth of cut. Increase in feed rate and depth of cut is found to increase the surface roughness.

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Keywords: Al6061; Surface Roughness; Turning Process; SN ratio; ANOVA

1. Introduction

Aluminium alloys are among the most commonly used lightweight metallic materials which possess attractive mechanical and thermal properties. Machinability of this material possesses relative easiness compared to other metals. Machinability quantifies the machining performance which can be defined by various criteria such as tool life, surface finish, material removal rate and power consumption (V. Songmene et. al, 2011), etc. Pure aluminium possess relatively low mechanical properties, which are improved by alloying the metal with other metals such as copper, manganese, silicon, magnesium, silicon, zinc, etc. Aluminium with magnesium and silicon as alloys are

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commercial designated as Al 6xxx series alloys. Among aluminium alloys Al 6061 exhibits excellent mechanical properties, corrosion resistance (Xuewu Lia,2015) and weldability (Omega Research,2002) which finds the applications in aircraft, missiles and space, ground and marine transportation, machine parts and architectural applications.

Nomenclature

A	Cutting Speed (m/min)
B	Feed (mm/rev)
C	Depth of Cut (mm)
Ra	Surface roughness μm
SN	Signal to Noise ratio
n	Total number of trial runs at i^{th} setting
y_i	Value of the response at i^{th} setting
k	Total number of replications.
DF	Degrees of freedom
SS	Sum of squares
MS	Mean sum of squares
F	Fisher's Number
d	Confidence interval
η_{eff}	Effective sample size

Toh et. al. (2004) studied the surface integrity effects on turned 6061 and 6061-T6 aluminium alloys. Investigations on machining of 6061 aluminium alloy with minimum quantity of lubricant, dry and flooded lubricant conditions were made by Sreejith (2008). Mukesh Kumar et. al. (2009) studied the effect of coated carbide tool on 6061-t4 aluminium. The effects of cutting parameters such as speed, depth of cut and cutting feed rate on the surface roughness were investigated by the authors and regression model are developed. Further, optimization of cutting parameters was carried by Carmita (2013) to minimize the energy consumption in turning of Al 6061-T6. Influence of cutting parameters on cutting force and surface finish was investigated by Rao et. al. (2013). It was found that feed rate is the significant factor that influence the cutting force as well as surface roughness. Also, depth of cut was found to have significant influence on the cutting force. Authors also have determined the optimum combination of feed rate and depth of cut to reduce power consumption as well as to achieve the better surface finish. Vinod Mishra et. al. (2014) studied the effects of tool overhang on selection of machining parameters and surface finish. It was found that too large and very small tool overhangs results in poor surface finish. Further, optimum range of tool overhang with minimum tool vibrations was established. Deepak et. al. (2015) studied the effect of coolant in turning of Al6061. It was found that use of coolant in machining improves the surface quality. In continuation of this, the present work attempts to optimize the process parameters and to study the interaction effect of the process parameters.

2. Methodology

Figure 1 shows the experimental set up used. Turning experiments were conducted using CNC turning center manufactured by ACE machine tool with Siemens control systems. The cutting tool used in the experimental work is silicon carbide insert. Material chosen in the present study is Al-6061 (as cast condition) is found to have tensile



strength of 140 MPa and hardness 60BHN. The process parameters considered in the present study are shown in the Table 1. The levels of these parameters are chosen based on the pilot study conducted at various settings. Experiments were designed using Taguchi method to use the experimental resources optimally. Experimental design is done with L_9 Taguchi orthogonal array. Experiments were conducted at various levels of the process parameters with two replications as shown in the Table 2. Figure 2 shows the sample work-piece machined. The surface roughness (R_a) obtained on the machined surface is measured using Taylor-Hobson Surtronic instrument. Table 2 also shows the R_a values obtained in each experimental trial.

Fig. 1 Experimental set up



Fig. 2 Sample work-piece

The experimental results were analyzed using Signal to Noise ratio with the Lower the Better criteria as given by the equation (1). The relative effect of each process parameter is analyzed by Analysis of Variance (ANOVA).

$$SN = -10 \log \left[\left(\frac{1}{n} \right) \sum_{i=1}^k y_i^2 \right] \quad (1)$$

Table 1 Signal parameter and their levels

Factor/Level	1	2	3
Cutting speed (m/min)	308	369	429
Feed rate (mm/rev)	0.05	0.1	0.15
Depth of cut (mm)	1	1.5	2

Table 2 Experimental plan and average results

A	B	C	R_{a1}	R_{a2}	Average R_a	SN Ratio
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308	0.05	1	0.7	1.01	0.855	1.22024
308	0.10	1.5	1.1	1.64	1.370	-2.8999
308	0.15	2	2.77	2.8	2.785	-8.8966
369	0.05	1.5	1.61	0.96	1.285	-2.4473
369	0.10	2	1.91	0.83	1.370	-3.3615
369	0.15	1	2.21	1.67	1.940	-5.8393
429	0.05	2	1.01	0.94	0.975	0.21431
429	0.10	1	0.66	1.22	0.940	0.16825
429	0.15	1.5	1.56	1.73	1.645	-4.3349

3. Results and Discussion

3.1 Effect of process parameters based on SN ratio analysis

The present work investigates the effect of operating parameters such as cutting speed, feed rate and depth of cut. Table 2 shows the surface roughness produced on the work-piece in each replication. To assess the variability of the response, the analysis is carried out by SN ratio using smaller the better criteria as the objective of the work is to minimize the surface roughness. The effect of each process parameter at different levels is calculated using equation (1) and is shown in Table 3. The relative effect of each factor on the response is shown in the form of deltastatistics in the same table. Based on the delta values the ranks are assigned to the factors in the ascending order of their influence.

Table 3. The mean SN ratio of process parameters

Level	Cutting Speed	Feed rate	Depth of cut
1	-3.5254	-0.3376	-1.4836
2	-3.8828	-2.0311	-3.2274
3	-1.3174	-6.3570	-4.0146
Delta	2.5653	6.0194	2.5310
Rank	2	1	3

Figure 3 shows the main effect plot of SN ratio for the surface roughness (R_a) as response. It is seen from the plot that lowest surface roughness is obtained at the process settings, cutting speed 429 m/min, Feed 0.05 mm/rev and depth of cut 1 mm. This combination of process parameter produces maximum SN ratio and hence the settings can be considered as optimum settings. Further, relative effect of each process parameter is determined by the ANOVA which is a statistical technique. In this technique the total variability of the response is separated with respect to individual contributions of each of the factors and the error. Table 4 shows the ANOVA values for the experimental results. The significance of effect of each factor on the response is evaluated using F test at 95 % confidence levels. It is seen that the effect of feed rate has highest influence on surface roughness followed by cutting speed and depth of cut. Also, the standardized residuals obtained in the ANOVA is shown in figure 4. It is observed from the figure that the residuals shows heteroscedasticity, which means experiments are conducted unbiased.

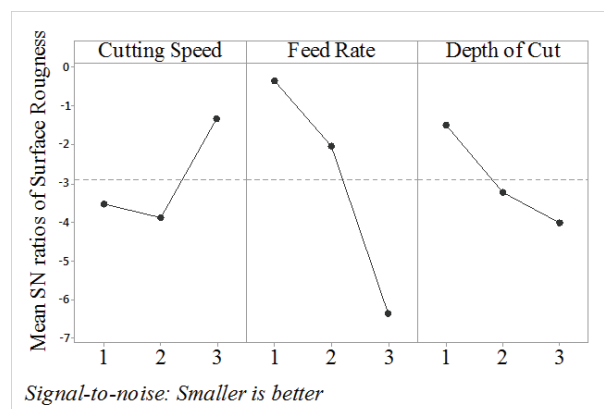


Fig. 3. Plot of SN ratio for the surface roughness

Table 4. ANOVA for the experimental data				
Source	DF	SS	MS	F
A	2	11.584	5.792	3.13
B	2	57.814	28.907	15.62
C	2	10.067	5.033	2.72
Error	2	3.701	1.850	
Total	8	83.165		

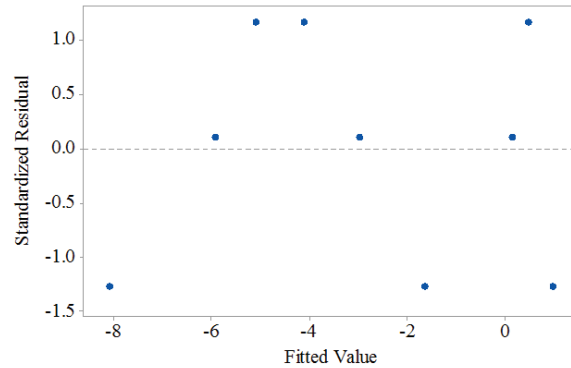


Fig. 4. The plot of standardized residual values

Further, Figure 5-7 shows the combined effect of process parameters on the surface roughness. As seen from the Figure 5, lower surface roughness is found to be produced at the settings of level 1 to feed rate and to the depth of cut. Further increase in these parameters increases the surface roughness drastically. The effect of depth of cut is seen to be highly significant compared to feed rate.

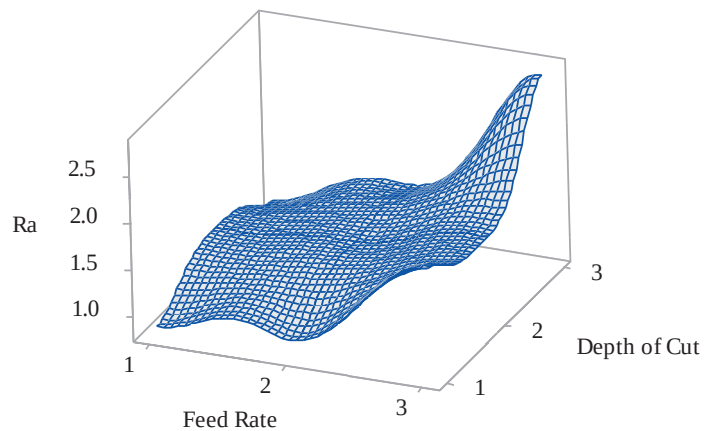


Fig. 5. The effect of feed rate and depth of cut on surface roughness

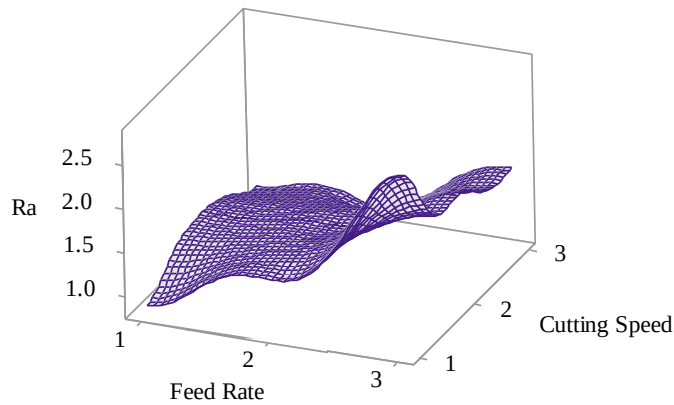


Fig. 6. The effect of feed rate and cutting speed on surface roughness

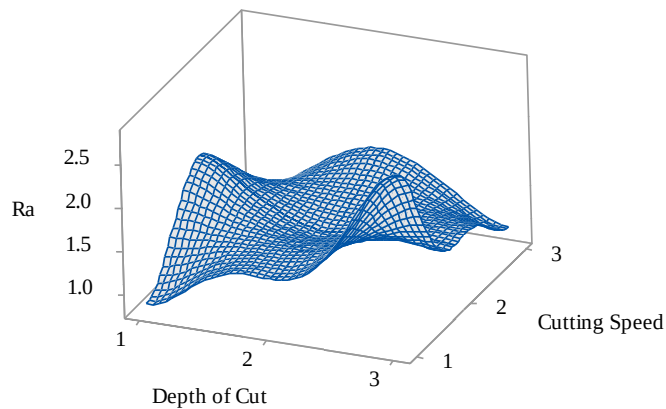


Fig. 7. The effect of cutting speed and depth of cut on surface roughness

Figure 6 shows the plot of surface roughness with respect to feed rate and cutting speed. It is seen that increase in the feed rate at any settings of cutting speed increases the surface roughness significantly. But it decreases with increase in cutting speed even at higher values of feed rate. Also, it is seen from the Figure 7 that the surface roughness exhibits more complex relationship with respect to the depth of cut and cutting speed. At the settings of the level 1 and level 3 respectively to these cutting parameters exhibits smoother cut surface. The surface roughness increases steadily with the increase in depth of cut. However, the corresponding increase in cutting speed along with depth of cut results in production of the smoother surface. The above phenomena is due to following facts: at higher cutting speeds more strain energy is available for the work-piece which leads to continuous and easy removal of the material thus producing smoother cut surface. The higher feed rates promotes the axial movement of the cutting tool resulting in shifting of the cutting tool to new location on the work surface without complete material removal and thus creates rough surfaces. Higher depth of cut creates higher thrust force on the cutting tool which push the tool radially outward. As a result the tool will be released from the work-piece causing vibrations which results in increase in surface roughness (Deepak *et. al.* (2015).

3.2 Prediction of surface roughness

The expected surface roughness (R_{ae}) at optimum settings of the process parameters i.e., at $A_3B_1C_1$ is calculated as below.

$$n_{eff} = \frac{\text{Total number of experiments}}{1 + \text{Total degree of freedom of all factors}} \quad (2)$$

$$\text{Average response (T)} = \frac{\sum \text{Response value}}{\text{Total number of experiments}} \quad (3)$$

$$R_{ae} = \text{Response at } [A_3 + B_1 + C_1] - 2T \quad (4)$$

$$d = \left[F(\alpha, \text{Error degree of freedom}) \frac{\text{Error mean square}}{\text{Effective sample size}} \right]^{\frac{1}{2}} \quad (5)$$

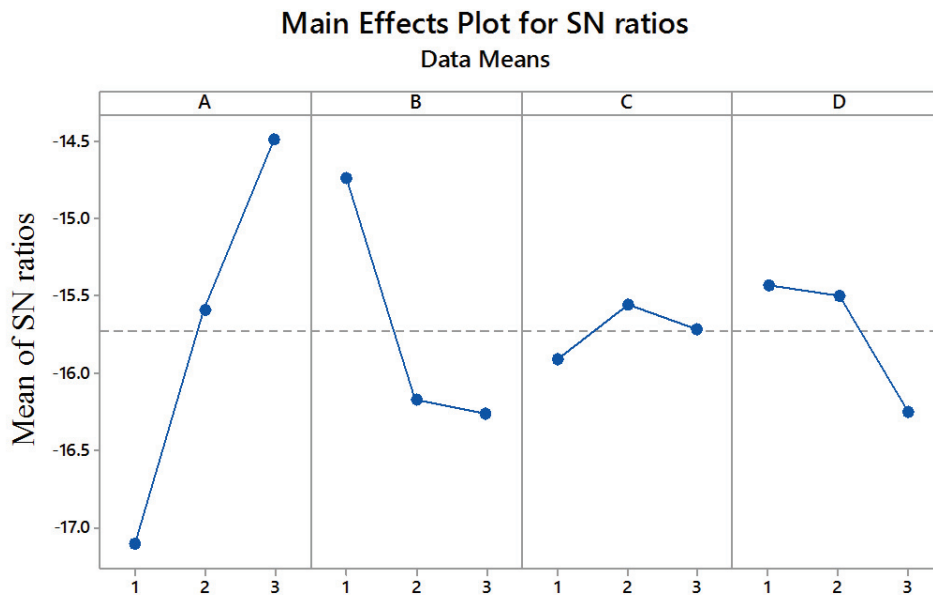
$$\text{Confidence interval} = R_{ae} \pm d \quad (6)$$

The confidence interval is estimated at 90 % confidence level. The expected surfaces roughness at optimum settings of process parameters is $0.438 \mu\text{m}$ with maximum deviation up-to $0.91 \mu\text{m}$. Further, confirmation experiments were conducted at the optimum settings. The surface roughness obtained at these is found within the range established.

4. Conclusion

Following conclusions are drawn from the present experimental work conducted using robust design principles of Taguchi.

- Feed rate is most influential process parameters that influence surface roughness while turning of Aluminum 6061



Signal-to-noise: Smaller is better